

Below is a **production-grade relational schema** optimized for:

- event intelligence
 - financial markets
 - AI prediction
 - user knowledge boards
 - high-volume time-series data
-

1. Database Architecture Overview

We will use **three data storage layers**:

1. Relational Core Database

Stores structured entities.

PostgreSQL

Contains:

- users
 - events
 - assets
 - boards
 - predictions
-

2. Time Series Database

Stores price data.

TimescaleDB

Contains:

- stock prices
 - crypto prices
 - commodity prices
-

3. Search Database

Elasticsearch

Used for:

- searching events
- searching news
- searching assets

2. Core Entity Model

Core relationship:

```
EVENT
  ↓
EVENT IMPACT
  ↓
ASSET
  ↓
PRICE DATA
  ↓
PREDICTIONS
```

Example:

Event:
Russia oil refinery attack

Assets:
Brent Crude
Exxon
Chevron

Prediction:
Oil ↑ 5%
Energy stocks ↑

3. Users Table

Stores platform users.

```
users
-----
id (UUID PK)
email
password_hash
username
role
created_at
updated_at
```

Optional:

```
subscription_plan
alert_preferences
```

4. Assets Table

Represents financial instruments.

```
assets
-----
id (UUID PK)
ticker
name
asset_type
sector
industry
country
exchange
currency
created_at
```

Example rows:

```
NVDA
TSLA
BTC
Gold
Brent Crude
```

Asset types:

```
stock
etf
commodity
crypto
forex
index
```

5. Events Table

Represents geopolitical or macroeconomic events.

```
events
-----
id (UUID PK)
title
description
event_type
country
region
severity
source
source_url
published_at
created_at
```

Example:

China bans gallium exports

Event types:

conflict
sanction
trade policy
economic data
energy disruption
election
regulation

Severity scale:

1-5

6. Event Tags Table

Allows flexible tagging.

```
event_tags
-----
id
event_id
tag
```

Example:

china
semiconductors
rare earth metals
trade war

7. Event Impact Table

This is **one of the most important tables**.

Maps events to assets.

```
event_impacts
-----
id
event_id
asset_id
impact_direction
impact_strength
confidence_score
created_at
```

Example:

```
event: Russia refinery attack
asset: Brent Crude
impact_direction: positive
impact_strength: 0.8
confidence: 0.65
```

Impact directions:

```
positive
negative
neutral
```

8. Predictions Table

Stores AI-generated predictions.

```
predictions
-----
id
event_id
asset_id
predicted_change
prediction_horizon
confidence_score
model_version
created_at
```

Example:

```
predicted_change = +4.3%
prediction_horizon = 24h
confidence = 0.67
```

Prediction horizons:

```
1h
6h
24h
7d
30d
```

9. Market Prices (Time-Series)

Stored in **TimescaleDB**.

```
market_prices
-----
id
asset_id
```

```
timestamp
open
high
low
close
volume
```

Example:

```
timestamp: 2026-03-12 10:30
asset: NVDA
price: 905.22
```

Indexes required:

```
asset_id
timestamp
```

10. News Articles Table

Stores raw news articles.

```
news_articles
-----
id
title
content
source
url
published_at
sentiment_score
created_at
```

11. Event Article Mapping

Links articles to events.

```
event_articles
-----
id
event_id
article_id
relevance_score
```

This allows **multiple articles per event**.

12. Boards Table (Pinterest Feature)

User boards.

```
boards
-----
id
user_id
title
description
visibility
created_at
```

Visibility:

```
public
private
```

13. Pins Table

Items saved to boards.

```
pins
----
id
board_id
content_type
content_id
created_at
```

Content types:

```
event
asset
prediction
news
```

14. User Watchlist

Tracks assets users follow.

```
watchlists
-----
id
user_id
asset_id
created_at
```

15. Alerts Table

Stores alert rules.

```
alerts
-----
id
user_id
asset_id
event_type
threshold
created_at
```

Example:

Alert:
Notify when energy-related events affect oil

16. Index Strategy

Indexes are critical.

Recommended:

```
events (published_at)
events (country)
events (event_type)

event_impacts (asset_id)

market_prices (asset_id, timestamp)

news_articles (published_at)
```

17. Example Query

Example:

Find assets affected by an event

```
SELECT
assets.ticker,
event_impacts.impact_direction,
event_impacts.confidence_score
FROM event_impacts
JOIN assets
ON assets.id = event_impacts.asset_id
WHERE event_impacts.event_id = 'EVENT123';
```

18. Example Data Flow

Pipeline:

News Article
↓
news_articles table
↓
NLP extracts event
↓
events table
↓
impact mapping
↓
event_impacts table
↓
AI prediction
↓
predictions table

19. Scaling Strategy

As the system grows:

Add:

partitioning for market_prices
read replicas
Redis caching

20. Strategic Insight

The **most valuable dataset** will be:

Event → Asset Impact → Market Reaction

Over time you will accumulate:

10k+ geopolitical events
100k+ asset reactions

This becomes a **proprietary geopolitical intelligence dataset**, which even hedge funds value.